Thanh Nguyen

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Contact Information

1 Hong Lac Str.
Tan Binh,
Ho Chi Minh City, Vietnam
Tel: (+84)396 939 805

Homepage: www.github.com
Linkedin: www.linkedin.com
✉ E-mail: www.gmail.com

Research Interests

- **Probabilistic machine learning and generative modeling:** theory and algorithms for diffusion models, score and flow matching, and optimal transport.
- **Robust learning under weak supervision:** noisy-label sequence modeling and positive–unlabeled learning for recommendation.

Education

Ho Chi Minh City University of Technology (HCMUT), Vietnam 2024–2026

- M.Res. in Applied Mathematics; GPA: **3.5/4.0** (8.42/10); ranked **1st** in cohort.
- Thesis: *Optimizing Score Flow Matching for Generative Models Based on Density Estimation* (grade **9.8/10**); advisor: Dr. Le Xuan Dai.

Dalat University: B.Sc. in Mathematics Education (second degree, part-time) 2024–2026

University of Economics HCMC: B.Sc. in Finance & Banking 2020–2023

Research Experience

- **AlignBC: Learning from Noisy Transcripts** Mar 2026–Present
  - Developed **AlignBC**, a training-only framework that adds alignment-weighted bounded costs to CTC-family objectives using posterior or unbalanced OT plans, without changing the decoder or inference.
  - Derived closed-form gradient updates and proved a mass–sensitivity factorization; achieved significant WER reductions in **7 of 8** CRNN-based STC/OTC comparisons on IAM and UIT-HWDB under synthetic transcript noise (five seeds; Holm-corrected tests).
- **Capacity-Dependent Theory for Kernel Score Estimation** Jan 2026–Aug 2026
  - Introduced a derivative-weighted effective dimension for kernel score estimation, giving rates $\widetilde{O}_P(M^{-(r+1/2)/(2r+1+\alpha)})$ beyond capacity-independent bounds.
  - Proved an exact derivative penalty $2/A$ for diagonal and curl-free torus kernels and a matching minimax lower bound, with guarantees holding at every diffusion time.
- **Graduate Research Assistant - Dr. Le Xuan Dai, HCMUT** Oct 2024–Mar 2026
  - *Generative models* (2025–2026): proposed objectives unifying KL, score and flow matching, and an OT-improved score flow matching objective (**2 journal papers**, M.Res. thesis).
  - *Bayesian optimization* (2024–2025): benchmarked BO for hyperparameter search.

Publications

- **Thanh Nguyen** et al. "AlignBC: Learning from Noisy Transcripts with Alignment-Weighted Bounded Costs." Submitted to IJCAI 2027.
- **Thanh Nguyen** et al. "Derivative-Weighted Capacity Bounds for Kernel Score Estimation." NeurIPS 2026 Workshop BeNTo.
- **Thanh Nguyen** et al. "Improving the Score Flow Matching Objective Function via Optimal Transport." *HUFLIT Journal of Science*, 2026.
- **Thanh Nguyen** et al. "Optimization of the Objective Function for Generative Models Based on Distribution Density Estimation." *HCMUE Journal of Science*, 2025.

Skills

- Python, PyTorch, JAX, Git, Linux, GPU Parallel, LaTeX; probability, optimization, ODEs, OT.

Additional Experience

- **Data Scientist Intern, PVCFC - PetroVietnam** Oct 2024–Apr 2025
- **International Mathematics Tutor** (IGCSE, IB, A-Level, Math olympiad) 2021–Present

References

Available upon request.